APMA 935 PROJECT: INVESTIGATING THE STABILITY OF A TRAFFIC LIGHT ON A RING ROAD

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This project will study numerical solutions to two conservative-hyperbolic traffic models, those being the Lighthill-Whitham-Richards (LWR) and Aw-Rascle-Zhang (ARZ) so called "second order" traffic model. The models are defined as follows.

1. Models. We note first the similarities between both traffic models to be examined in this project, that is that they model cars as a density continuum, rather than a system of discrete cars. Due to this conceit, these models are to be understood as a wide perspective on large traffic patterns over such distance that the decisions of individual cars blur into the behavior of a continuum–of fluid.

We take as common quantities between the two traffic models a density ρ and a velocity v. As this is a mathematical project first and foremost the problem will be scaled such that $0 \leq \rho \leq 1$ and $0 \leq v \leq 1$. Finally, traffic is one of the few topics which is naturally modeled by strictly one dimensional equations so long as we limit our ambitions to such phenomenon as highway travel lacking intersections, and other such forays into the second dimension.

1.1. Lighthill-Whitham-Richards Traffic. We begin with the celebrated Lighthill-Witham-Richards (LWR) traffic model derived from two basic assumptions. The first is a the natural conservation law of one dimensional traffic flow: "No Passing". Mathematically we can write that the total integrated density between two points $x_1(t)$,$x_2(t)$ advected by the flow (velocity) must remain constant. To do otherwise would imply a car (or some amount of car-modeling fluid) has passed the car in front of it, or been passed from behind. We write:

$$\frac{d}{dt}\int_{x_1(t)}^{x_2(t)} \rho(x,t)dx = 0. \tag{1.1}$$

The second assumption is that velocity is a function of density, $v(\rho)$, that drivers would be cautious and slower in a packed rush hour and contrastingly free and expeditious when nearly alone. To this end we assume that $v(\rho)$ is a monotonically decreasing function. Now, expanding 1.1 using Leibniz' rule we arrive at the conservative form of LWR

$$\partial_t \rho + \partial_x \left(\rho v(\rho)\right) = 0. \tag{1.2}$$

Equivalently we define an overall "flux function" $q(\rho) = \rho v(\rho)$ which describes the throughput of traffic and write:

$$\partial_t \rho + \partial_x q(\rho) = 0 \tag{1.3}$$

For the purpose of this project we will consider $v(\rho) = 1 - \rho$ as the explicit relationship between velocity and density. In particular the reader is encouraged to notice that this leads to two scenarios of zero flux $q(\rho) = 0$ at $\rho = 1$ (Gridlock) and $\rho = 0$ (Vaccuum).

This manuscript is for review purposes only.

1.2. Aw-Rascle-Zhang Traffic.

1.2.1. Brief History. Aw-Rascle-Zhang (ARZ) traffic we believe is best understood as a small step up in complexity from the–in Aw & Rascles [1] own words–"celebrated" LWR model, celebrated perhaps for its elegance and perhaps even usefulness in demonstrating the possibilities of continuum based traffic models, however we believe it is easy to see that the LWR model's simplicity is as much of a curse as a blessing.

Daganzo [2] is much in agreement noting the LWR models inability to represent the dispersive properties of traffic flow caused by the different speeds enjoyed by different drivers allowing faster drivers to pull away and slower drivers to fall behind as happens on multilane highways.

Additionally it can be shown that a shock will form under the LWR model possessing points $x_1 < x_2$ such that $\rho(x_1) < \rho(x_2)$ which in our opinion limits the models ability to model situations such as vehicle collisions–LWR appears to believe that collisions (that is to say, shocks and thus instantaneous changes in velocity) are the only way in which a vehicle can slow down.

Daganzo continues his paper in critique of the therefore proposed second order traffic models. It is however in 2000 that Aw & Rascle publish a near direct response, incorporating Daganzo's criticism and emerging with the topic of our section: The ARZ traffic model.

1.2.2. Aw & Rascle Traffic. We begin with the same starting points as LWR requiring that cars as a density is conserved by the differential equations. This is the first equation of Aw & Rascle traffic:

$$\partial_t \rho + \partial_x (\rho v) = 0. \tag{1.4}$$

However we now allow v to vary as a quantity which is responsive to ρ but not as an explicit function of density. Instead we define a "pressure" function $p(\rho)$ in analogy to fluid mechanics as an increasing function of density. Then we add a second coupled equation

$$\partial_t (v + p(\rho)) + v\partial_x(v + p(\rho)) = 0. \tag{1.5}$$

The reader is encouraged to recognize the presents of the "advective derivative", $\frac{D}{Dt}(\dots) = \partial_t(\dots) + v\partial_x(\dots)$ which describes rate of change along particle paths. This new equation can be rearranged to show the underlying meaning of this second equation.

$$\frac{D}{Dt}(v + p(\rho)) = -|p'(\rho)|\frac{D}{Dt}\rho. \tag{1.6}$$

This equation shows that velocity along a particle path, a car path, if you will, that an increase in density directly results in a reduced velocity. What Aw & Rascle refer to as *driver anticipation*.

1.2.3. Zhang Traffic. Missing so far from the discussion of ARZ traffic is its final letter, Zhang. Working independently from Aw & Rascle and in dialogue with the very same Daganzo [2] article, developed a similar traffic model to combat the

inequities of LWR focusing around the idea of an *equilibrium velocity* $V_e(\rho)$ analogous to the LWR $v(\rho)$ relation.

Conservation of mass is shared between the Aw, Rascle and Zhang perspectives, the velocity equation for Zhang traffic can written

$$\partial_t \left(v - V_e(\rho)\right) + v\partial_x(v - V_e(\rho)) = 0. \tag{1.7}$$

which additionally shows why the Aw & Rascle model and the Zhang model are often considered together as separate perspectives on the same system. Indeed, choosing $p(\rho) = v_{max} - V_e(\rho)$ makes both models identical.

The Zhang perspective is the one used most prominently in this project for the purpose of interpretation of results.

1.2.4. Relaxation. A consequence that becomes clear at the appraisal of 1.7 is that for an individual particle path, that is to say, an individual driver, we expect whatever disagreement of said driver with respect to the equilibrium (lawful?) traffic speed will be conserved along the path of their car! Professor Muraki pointed out this interpretation along with the moniker "drivers with attitude" either timid or antisocial attitudes respectively. This perspective will be experimented on in the numerical results section.

Another perspective is that equilibrium velocity is to be understood as a universally comfortable driving speed, and the desination velocity behavior of all drivers on the road. To model this perspective we have the ability to include a so called *relaxation term* to our velocity equation

$$\partial_t \left(v - V_e(\rho)\right) + v\partial_x(v - V_e(\rho)) = A(V_e(\rho) - v) \tag{1.8}$$

to correct this deviance over time.

Similarly to the LWR model we will describe $V_e(\rho) = 1 - \rho$. Important to note, that the tendency of this model with relaxation is to relax any perturbations in velocity to the point where the velocity solution v is indistinguishable from LWR traffic.

2. Numerical Methods. These PDE's will be solved using conservative finite volumes approaches to solve for discontinuous weak solutions to the PDES. In particular we will be using an exact Godunov approach for both models, as well as Roe's matrix approximation for LWR and presenting a Roe approximation for ARZ which has not been implemented.

In general, both approaches seek to solve a conservative hyperbolic PDE on some domain $[a, b] \times [0, T]$ which can be written in general form as

$$\partial_t \vec{u} + \partial_x \vec{q}(\vec{u}) = 0. \tag{2.1}$$

Where $\vec{q}$ is a function describing the flux at some solution state.

We discretize the solution into M equally sized cells U_j^n for $j = 1, \ldots, M$ such that $\vec{u}(x_j, t_n) = u(a + j\Delta x, n\Delta t) \approx U_j^n$. Also of importance are the interfaces of each cell at $x_{j+1/2}$. Each cell U_j can be imagined as the average value that an approximate solution takes over $[x_{j-1/2}, x_{j+1/2}]$.

2.1. Exact Godunov. The exact Godunov numerical solve requires an exact solution to the Riemann problem created by the discontinuity at the boundary between two adjacent cells. In particular, over some time step Δt we wish to calculate

the exact flux at each interface of each cell. As notation, we refer to the flux between cells U_j^n and U_{j+1}^n as

$$\vec{q}\left(\vec{u}(x_{j+1/2}, t_n)\right) = Q(U_j^n, U_{j+1}^n)$$

A Godunov timestep then increments each cell based on the exact flux across its boundaries written as

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} = \frac{1}{\Delta x}\left(Q(U_{j-1}^n, U_j^n) - Q(U_j^n, U_{j+1}^n)\right)$$

or

$$U_j^{n+1} = U_j^n + \frac{\Delta t}{\Delta x}\left(Q(U_{j-1}^n, U_j^n) - Q(U_j^n, U_{j+1}^n)\right)$$

This may sound like a relatively difficult process, however we can utilize a core property of hyperbolic PDE's to simplify the method considerably. Described in much more detail in LeVeque [3], we can calculate the maximum wave speed in the hyperbolic system $|\lambda_{\max}|$ and choose our time discretization no waves reach far enough into the cells to interact with other waves from other interfaces and therefore, the exact flux at the cell interfaces are constant for Δt satisfying a CFL condition. This CFL condition can be written

$$\left|\frac{\Delta t}{2\Delta x}\lambda_{\max}\right| \leq 1,$$

and allows us to calculate an optimum timestep

$$\Delta t = \frac{2\Delta x}{|\lambda_{\max}|} * (1 - \mathrm{TOL}) \tag{2.2}$$

for some decided percentage tolerance.

Without detailing the complete analysis of each hyperbolic system, it can be shown that trivially, the maximum wave speed of the LWR system is $v_{max} = 1$ for this project. The wavespeeds of the ARZ system are calculated by Mammar et al. [4] as

$$\lambda_1 = v + \rho V_e'(\rho) \qquad \lambda_2 = v.$$

Since $V_e(\rho)$ is a decreasing function, we find that v is again the largest wavespeed and thus $\lambda_{max} = v_{max} = 1$

2.2. Roe Solver. The second numerical method employed is using Roe approximation matrices, the logic goes that instead of solving the exact Riemann problem created between two cells, we derive a linear approximation of the system at each interface and solve the Riemann problem of this linear approximation. We write this approximation of the hyperbolic conservation law 2.1 as a linear equation

$$\partial_t \vec{u} + [A]\partial_x \vec{u} = 0. \tag{2.3}$$

The difficulty in this approach comes from the discontinuity itself, we are deprived at the interface of the Taylor series as a tool and must be very clever in choosing our linear approximation. Someone who was very clever was Roe [5], who proposed a list of properties that this $[A]$ must posses to be a "good" approximation.

i) As $U_{j+1} \to U_j \to \vec{u}(x_{j+1/2}, t)$ we need $[A] \to J\left(\vec{u}(x_{j+1/2}, t)\right)$ where $J(\vec{u})$ is the Jacobian of $\vec{q}$.

ii) We need $[A](U_j - U_{j+1}) = q(U_j) - q(U_{j+1})$. This is the Rankine-Hugoniot condition.

iii) The eigenvectors of $[A]$ must be linearly independent.

We can now use this simplified system to easily calculate the requisite fluxes and increment the numerical solution–except, there is a detail. This linear approximation is unable to represent a class of Riemann problem solutions known as *rarefaction waves*. The solution to this problem involves an *entropy fix* to ensure that we remain on a particular desirable weak solution to the PDE and avoid problems of non-uniqueness.

For a system of one conservation law, the Roe matrix can be solved for explicitly for a general hyperbolic conservation law:

$$\partial_t u + \partial_x q(u) = 0. \tag{2.4}$$

First, assuming an initial Riemann problem defined by u_L and u_R we can solve for ii) explicitly,

$$[A] = \frac{q(u_L) - q(u_R)}{u_L - u_R} \tag{2.5}$$

and notice that this satisfies i) by the difference quotient and iii) trivially. Thus our Roe "matrix" for LWR is simply.

$$[A_{\text{LWR}}] = 1 - 2\left(\frac{\rho_L + \rho_R}{2}\right) = 1 - 2\tilde{\rho} \tag{2.6}$$

The ARZ system is more resistant to Roe approximation, however, the following Roe approximation has been calculated and verified using the black box of symbolic calculators (Sympy). Unfortunately a "Roe Average" notation for this matrix has not been found.

$$
\begin{aligned}
A_{00} &= 1 - 2\left(\frac{\rho_L + \rho_R}{2}\right) \\
A_{01} &= 1 \\
A_{10} &= -\frac{y_L + y_R}{2} - \frac{(y_L - y_R)^2 + \frac{(\rho_L y_R - \rho_R y_L)^2}{\rho_L \rho_R}}{(\rho_L - \rho_R)^2} \\
&\quad + \frac{2(y_L - y_R)(\rho_L y_R - \rho_R y_L)}{(\rho_L - \rho_R)^3} \ln\left(\frac{\rho_R}{\rho_L}\right) \\
A_{11} &= 1 - \frac{\rho_L + \rho_R}{2} + \frac{2(y_L - y_R)}{\rho_L - \rho_R} + \frac{2(\rho_L y_R - \rho_R y_L)}{(\rho_L - \rho_R)^2} \ln\left(\frac{\rho_L}{\rho_R}\right)
\end{aligned}
$$

$$
[A_{\mathrm{ARZ}}] = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \tag{2.7}
$$

Now without derivation, the required flux at an interface for a linear system of m conservation laws is given by a eigenvector sum. First, we represent the left and right states as a combination of eigenvectors

$$
U_{j-1}^n = \sum_{j=1}^{m} \alpha_j \vec{v}_j \qquad U_j^n = \sum_{j=1}^{m} \beta_j \vec{v}_j
$$

the define the following "jump vector"

$$
\sum_{j=1}^{m} (\beta_j - \alpha_j) \lambda_j \vec{v}_j.
$$

And the Godunov flux is then written

$$
Q(U_{j-1}^n, U_j^n) = [A] U_{j-1}^n + \sum_{j \text{ s.t } \lambda_j < 0} (\beta_j - \alpha_j) \lambda_j \vec{v}_j.
$$

This matrix will unfortunately go unused for this project, but we figured it might be interesting to the reader.

2.2.1. The Entropy Fix. A detail to resolve with the Roe approximation method is that a linear hyperbolic conservation law cannot accurately represent a rarefaction wave requiring a so called "entropy fix" to ensure that the linearized system converges to the correct entropy solution. For the LWR Roe approximation we will be making use of the Harten-Hyman entropy fix as described in LeVeque [3]. For a detected sonic rarefaction, we modify the flux

$$
Q(U_{j-1}^n, U_j^n) = q(U_{j-1}^n) + q'(U_{j-1}^n) \left(\frac{q'(U_j^n) - s}{q'(U_j^n) - q'(U_{j-1}^n)} \right) (U_j^n - U_{j-1}^n). \tag{2.8}
$$

This is perhaps not worth the effort or computation for such a simple system, but is performed here to allow for benchmarking the Roe approximation for LWR as though we were not in possession of an exact Riemann problem resolution.

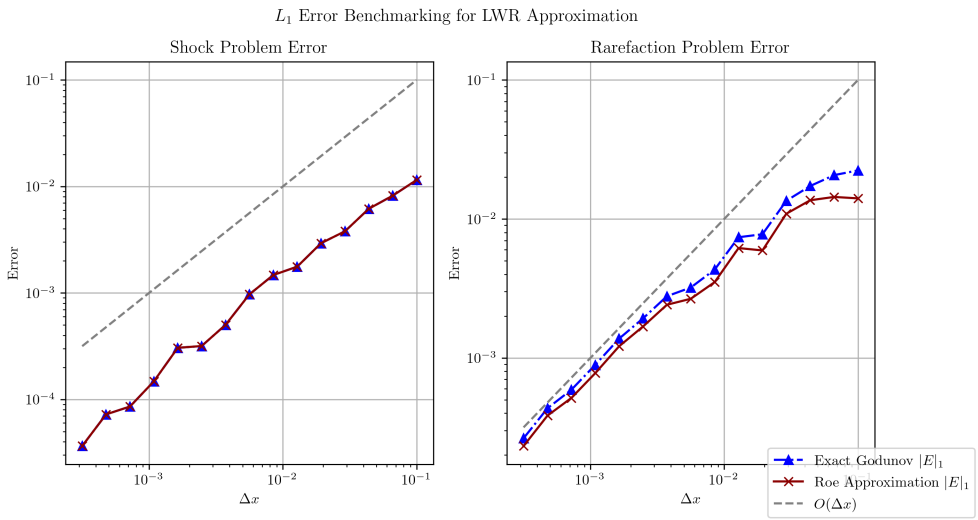

Fig. 1: LWR error results at $t = 1$

2.3. Convergence Study.

2.3.1. Lighthill-Whitham-Richards. We are blessed for the convergence study of LWR to be able to produce exact solutions to the Riemann problem with which to benchmark our solver. In particular, two problems will be computed and compared to exact solutions, the shock causing Riemann problem and the rarefaction causing Riemann problem respectively.

Consider:

$$\partial_t \rho + \partial_x(\rho(1-\rho)) = 0 \qquad (x,t) \in [-\infty, \infty] \times [0,1] \tag{2.9}$$

$$\rho(x,0) = \begin{cases} \rho_L, & \text{if } x < 0 \\ \rho_R, & \text{if } x > 0 \end{cases}$$

Where the shock problem takes $\rho_L = 0.3$ and $\rho_R = 0.5$ and the rarefaction problem takes $\rho_L = 0.5$ and $\rho_R = 0.2$. We approximate free flow boundary conditions on our numerical tests via a quadratic interpolation.

Figure 1 shows the results of our convergence tests clearly demonstrating the $O(\Delta x)$ error expected from our simple Gudonov scheme. In addition, we notice exact agreement in the case of the shock test where the Roe approximation method is equivalent to exact Godunov, whereas the rarefaction test shows the effect of the approximate entropy fix. Interestingly however, we note that approximation gives superior error to the exact method! We suspect this is due to chance cancellation between upwinding error and entropy fix error.

2.3.2. Aw-Rascle-Zhang. The ARZ traffic model is less straightforward to verify. Lacking exact solutions we will present two results to show convergence. First we will demonstrate a point wise convergence to one particular function (we hope

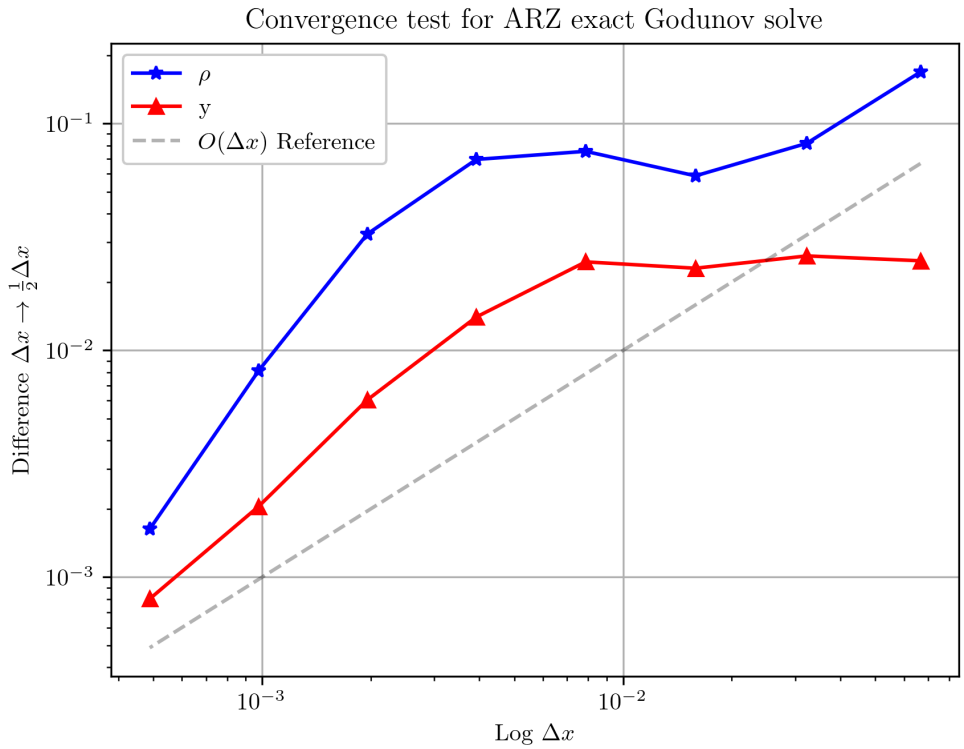

Fig. 2: L_2 norm of difference between common points of sequentially refined discretizations.

this is the solution), then we demonstrate that the numerical method preserves the conserved quantities of the system: ρ, y where $y = \rho(v + p(\rho))$.

We again perform our study on the Riemann problem, this time for the ARZ system, this time with periodic boundary conditions.

$$
\begin{aligned}
\partial_t \rho + \partial_x(\rho(1-\rho)) &= 0 \qquad (x,t) \in [0,1] \times [0,1] \\
\text{(No realxation)} \qquad \partial_t(v + p(\rho) + v\partial_x(v + p(\rho)) &= 0 \\
\rho(0,t) &= \rho(1,t) \\
v(0,t) &= v(1,t)
\end{aligned}
$$

$$
\rho(x,0) = \begin{cases} \rho_L, & \text{if } x < 0 \\ \rho_R, & \text{if } x > 0 \end{cases}
$$

$$
v(x,0) = \begin{cases} v_L, & \text{if } x < 0 \\ v_R, & \text{if } x > 0 \end{cases}
$$

For our pointwise convergence study we begin by finding particular N (number of cells) such that the resulting discretizations share a number of points $\{x_{\text{common}}\}$.

The particular N used are [7, 15, 31, 63, 127, 255, 511, 1023, 2047] which all share the points $x = \{0.125, 0.250, 0.375, 0.500, 0.625, 0.750, 0.875\}$. We provide evidence of convergence by taking the sequence of solutions $\{\vec{u}_{N_j}\}$ and taking plotting the normed of the difference between sequential solutions $||\vec{u}_{N_j} - \vec{u}_{N_{j-1}}||_2$. The results of this study on the Riemann problem with $(\rho_L, v_L) = (0.2, 0.5)$ and $(\rho_R, v_R) = (0.3, 0.3)$ are seen in figure 2.

Additionally, as this is meant to be a conservative system, we can benchmark our method's ability to conserve the required quantities. Using the same Riemann initial data as the pointwise convergence study we note the value of the integral

$$\text{Conservation Error} = \int_0^1 \begin{bmatrix} \rho(x,0) \\ v(x,0) \end{bmatrix} dx - \int_0^1 \begin{bmatrix} \rho(x,T) \\ v(x,T) \end{bmatrix} dx$$

which records the total amount of each conserved quantity. Then for increasingly refined grids, we compute the same integral at $t = 1$. An exact solution should keep both measurements equal, so the difference between our measurements at $t = 0$ and $t = 1$ will demonstrate an error in our numerical solution. From this study we can confirm that indeed our conserved quantities are for the most part respected by our numerical methods (figure 3)

3. Red Light Green Light. As the title suggests the principle concern is to use the ARZ and LWR models to investigate the problem of a traffic light on a circular "ring road". Since ARZ with relaxation tends towards LWR, for the purpose of analyzing stability we will be performing our numerical analysis of stability using LWR exclusively. The possibility of unstable non-relaxed ARZ will be discussed briefly in a later section. The problem can be stated thusly:

$$\partial_t \rho + \partial_x(\rho(1-\rho)) = 0 \qquad (x,t) \in [0,1] \times [0,T]$$
$$\rho(x,0) = \rho_0$$

With boundary conditions based on a parameter τ representing half the period of the traffic light. For $t \in [0, \tau]$ the light is red. This proscribes the no-flux boundary condition on the right hand side (there is no need for a left hand side boundary condition as flow is unidirectional).

$$\rho(0,t) = 0$$
$$\rho(1,t) = 1$$

Notice that both $\rho = 0, 1$ results in zero flux. Then for $t \in [\tau, 2\tau]$ we proscribe the green-light boundary condition

$$\rho(0,t) = \rho(1,t)$$

that is to say, periodic. Then for $t \in [2\tau, 3\tau]$ we have another red light, then green light, and continuing in this fashion. We define a stable solution in this arrangement as a solution which tends towards a density function which is periodic in time.

Useful for this problem is an understanding of the precise shockspeed formula for the LWR system obtained by the Rakin-Hugoniot condition:

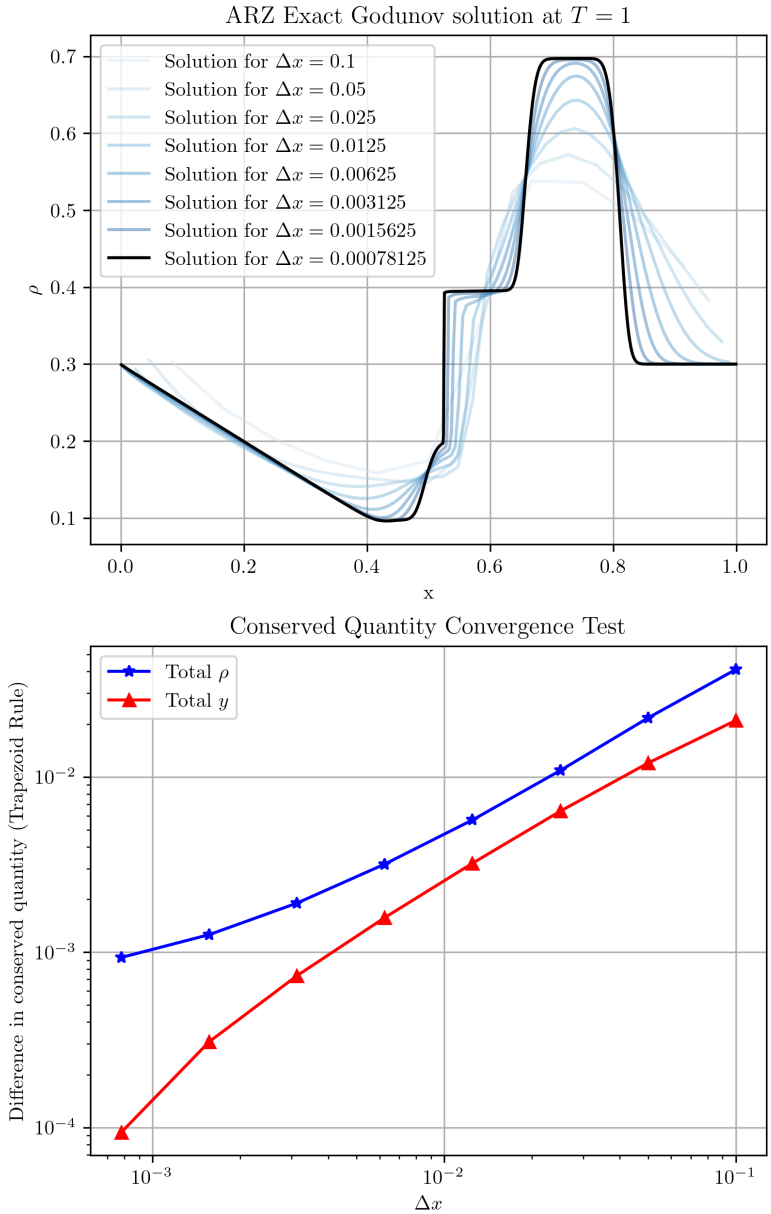

Fig. 3: (Left) Computed solution plot at $T = 1$ for various N. (Right) Log Log error plot of conservation error vs Δx

$$s = 1 - 2\left(\frac{\rho_L + \rho_R}{2}\right) = 1 - \rho_L - \rho_R \tag{3.1}$$

where s is the shock speed.

In figure 4 we can see a representative snapshot of this problem taken after the

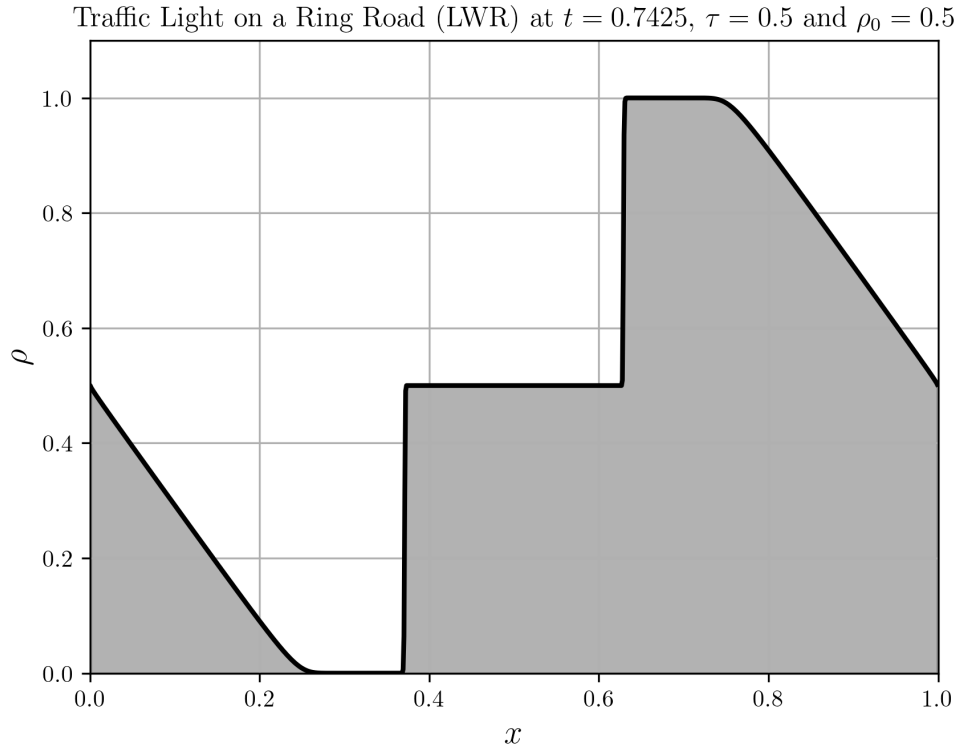

Fig. 4: Snap shot of the traffic light problem

light has turned green and allowed a glut of jammed traffic to be begin to clear. Of note are the two shocks and rarefaction which straddles the periodic boundary. The right most shock has $s = -0.5$, this is the remaining cars waiting to accelerate through the light. The left most shock has $s = 0.5$ representing the last car who cleared the previous red light after which no cars were allowed to flow. The rarefaction represents cars beginning to accelerate through the traffic light.

We can additionally visualize this system via a *shock plot* as seen in figure 5. Which plots the location of shocks in the simulation. We can see a consistent pattern of shocks originating on the left and right hand side of the domain and traveling at equal speed to the center where they merge into a central shock which lasts from creation until the end of the simulation. It is this style of plot which will be used to suggest the stability of the solution.

We can modify τ and ρ_0 in order to see the variance of possible shock behavior as seen in figure **??**

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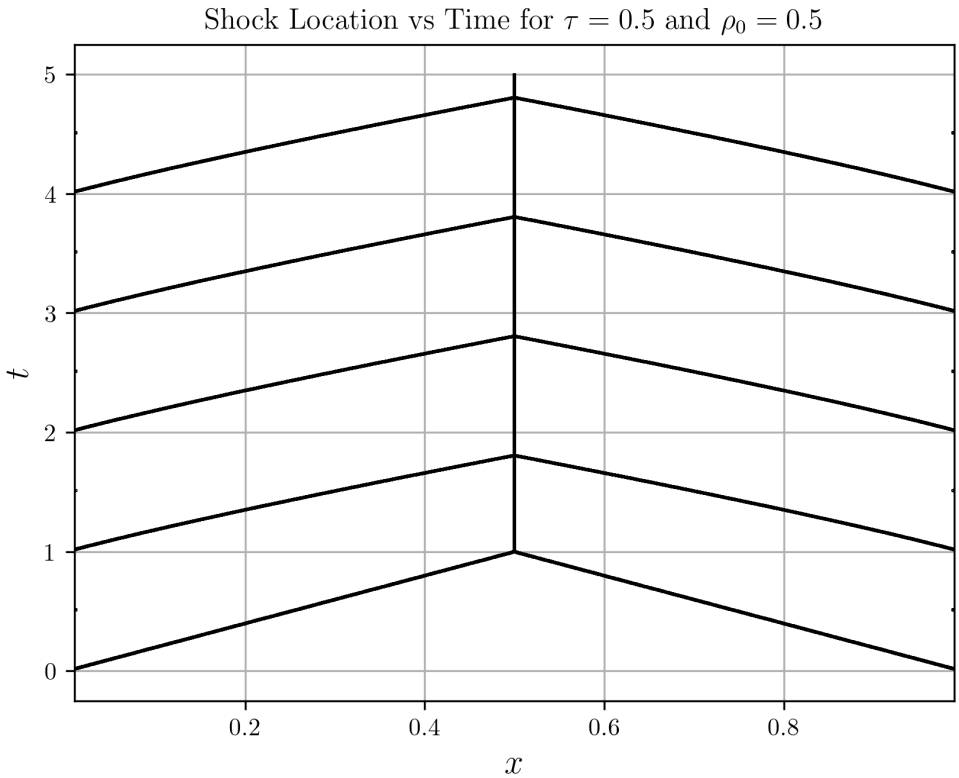

Fig. 5: Shock movement for the ring road traffic light problem with $\tau = 0.5$

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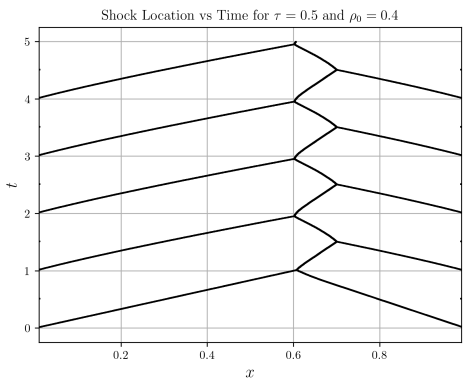

Figure 1

figure2.jpg

Figure 2

figure3.jpg

Figure 3

figure4.jpg

Figure 4

Fig. 6: Main caption for the 2×2 figure grid